

The two- $\mathfrak{C}$ row

Checked, regulated, and the $+$ prescription answered

$$S = p^+ + k^+, \zeta = k^+/S, \bar{\zeta} = p^+/S, r = y' - y; \quad X^k = \frac{(x-y)^k}{(x-y)^2}, X'^{k'} = \frac{(x'-y')^{k'}}{(x'-y')^2}; \quad \mathcal{X} = X \cdot X', \mathcal{R} = (X \cdot \hat{r})(X' \cdot \hat{r})$$

The calculation is right. Prefactor ratio exactly 1; the w, w' integration including the subtle cancellation between the Jacobian and the kernel rescaling; the phase; and all three tensor structures of your post-integration bracket, each verified independently (§1–§3). The Wilson–line reduction to $N_c[U(y') - U(x')][U(y) - U(x)]$ is **the right answer**, but your index orders have to be made uniform first — as written the reduction does not go through (§4). **On divergences: there is exactly one**, at $|z| \rightarrow \infty$, and its residue is *exactly* $\frac{P_{gg}(\zeta)}{2N_c k^+} \mathcal{X}$ — a **collinear** pole, which is what makes this the fragmentation row (§5). No UV anywhere. After $\overline{\text{MS}}$ (§6) the row is

$$\frac{\pi}{k^{+2}} \left\{ C_{\text{UV}}(\zeta) \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\},$$

and **yes, you may use the $+$ prescription here** — unlike in the $|\mathfrak{B}_2|^2$ row — for a reason that is now checkable (§8).

1. Prefactor — exact

With $\mathfrak{A}^{(1)}$ evaluated at $p^+ + k^+$ and $\mathfrak{C}_1$ at $k_{\text{arg}}^+ = S$ (so $\sqrt{p^+(S - p^+)}/S = \sqrt{p^+ k^+}/S$):

$$|\mathfrak{A}^{(1)}|^2 = \frac{g^2}{2\pi^2 S}, \quad |\mathfrak{C}_1|^2 = \frac{g^2 p^+ k^+}{8\pi^2 S^3}, \quad \frac{4}{(2\pi)^3} |\mathfrak{A}^{(1)}|^2 |\mathfrak{C}_1|^2 = \frac{g^4}{32\pi^7} \frac{p^+ k^+}{S^4}$$

= your $\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{p^+ k^+}{S^4}$, ratio 1. The four phases $(-i)(-1)(-1)(+i) = 1$, so the row is real.

The $\Theta(k_{\text{arg}}^+ - p^+ - \Lambda)$ becomes $\Theta(k^+ - \Lambda)$ here, so it imposes nothing on p^+ — the limits are the ones you wrote.

2. The w, w' integration — and the cancellation that makes it work

$\delta^{(2)}[y - w + \frac{p^+}{S}(w - z)]$ has w -coefficient $-\frac{k^+}{S}$, so

$$w = \frac{Sy - p^+ z}{k^+}, \quad \int d^2 w \delta^{(2)} = \left(\frac{S}{k^+}\right)^2 \text{ per delta} \Rightarrow \left(\frac{S}{k^+}\right)^4 \text{ for both.}$$

But the $\mathfrak{C}$ kernel is **not** inert under the substitution: $w - z = \frac{S}{k^+}(y - z)$, so

$$\frac{(w - z)^m}{(w - z)^2} = \frac{k^+}{S} \frac{(y - z)^m}{(y - z)^2}, \quad \text{two of them} \Rightarrow \left(\frac{k^+}{S}\right)^2.$$

$$\text{net factor} = \left(\frac{S}{k^+}\right)^4 \times \left(\frac{k^+}{S}\right)^2 = \frac{S^2}{k^{+2}} \text{ — which is what you used.}$$

(Worth flagging because using $\frac{S^4}{k^{+4}}$ alone, or forgetting the Jacobian entirely, both give the wrong answer; only the combination is right.) And

$$w' - w = \frac{S}{k^+}(y' - y) \implies e^{-ik(w' - w)} = e^{-i\frac{k}{\zeta}(y' - y)} = e^{-ik \cdot r} e^{-i\frac{p^+}{k^+} k \cdot r},$$

exactly your split. **This is the fragmentation signature:** the hard factor is evaluated at the *parent's* transverse momentum $k_{\perp}/\zeta$, not at $k_{\perp}$.

3. The tensor contraction — all three structures exact

First, an index note: $\mathfrak{C}_{1jki}$ is defined with order (j, k, i) but called in the row as $\bar{\mathfrak{C}}_{1jik}$, order (j, i, k) . Swapping $k \leftrightarrow i$ in the definition's bracket turns $\delta_{im}\delta_{jk} - \frac{S}{p^+}\delta_{jm}\delta_{ik} - \frac{S}{k^+}\delta_{km}\delta_{ij}$ into exactly what you wrote. Correct — just make the relabelling explicit.

Contracting the two brackets over i, j gives, for every p^+, k^+ tried (agreement 10^{-15}):

$$c_1 \delta_{km} \delta_{k'm'} + c_2 \delta_{mm'} \delta_{kk'} + c_3 \delta_{mk'} \delta_{km'}, \quad c_1 = d_\perp - \frac{2S}{p^+} - \frac{2S}{k^+}, \quad c_2 = S^2 \left[\frac{1}{p^{+2}} + \frac{1}{k^{+2}} \right], \quad c_3 = \frac{2S^2}{p^+ k^+}.$$

Multiplying by (prefactor) $\times$ (net w factor) $\times k^+ = \frac{p^+ k^+}{S^4} \cdot \frac{S^2}{k^{+2}} \cdot k^+ = \frac{p^+}{S^2}$:

structure	computed	yours	difference
$\delta_{km} \delta_{k'm'}$	$\frac{d_\perp p^+}{S^2} - \frac{2}{k^+}$	same	0
$\delta_{mm'} \delta_{kk'}$	$\frac{p^+}{k^{+2}} + \frac{1}{p^+}$	same	0
$\delta_{mk'} \delta_{km'}$	$\frac{2}{k^+}$	same	0

Note the $-\frac{2}{k^+}$ in the first came from $-\frac{2S}{p^+} - \frac{2S}{k^+}$ collapsing by $\zeta + \bar{\zeta} = 1$; your grouping $\frac{2}{k^+}(\delta_{mk'} \delta_{m'k} - \delta_{m'k'} \delta_{km})$ is exactly right.

4. The Wilson–line reduction — right answer, wrong index orders

With uniform orderings, i.e.

$$\text{ket} = f^{cda} U^{db}(z) [U^{ec}(y) - U^{ec}(x)] \rho^e(x), \quad \text{bra} = f^{c'd'a} U^{d'b}(z) [U^{e'e'}(y') - U^{e'e'}(x')] \rho^{e'}(x'),$$

the reduction runs $\sum_b U^{d'b}(z) U^{db}(z) = \delta^{d'd}$, then $\sum_{a,d} f^{c'da} f^{cda} = N_c \delta^{c'c}$, giving $N_c [U(y') - U(x')] [U(y) - U(x)]$ — verified to $3 \cdot 10^{-15}$. **Your quoted answer is correct.**

But as literally written it does not follow — the discrepancy is 5.11 in norm. Two problems, both the same family:

1. **The z lines are transposed between the two terms:** $U^{db}(z)$ in term 1 versus $U^{bd}(z)$ in term 2 (and $U^{d'b}$ vs $U^{bd'}$ in the bra). Then the cross terms give $\sum_b U^{d'b} U^{bd} = [U(z)U(z)]^{d'd}$, which is not a delta, so no N_c . Your **top–line** expression has $U^{bd}(z)$ in *both* ket terms, which is consistent — the expanded version is where it broke.
2. **The y and x lines are transposed:** $U^{ec}(y)$ versus $U^{ce}(x)$. This one matters more than bookkeeping: $[U(y) - U(x)^T]$ does **not** vanish at $y = x$ ($\|U(x) - U(x)^T\| = 1.24$), and that vanishing is exactly what tames the $y \rightarrow x$ end of the $\mathfrak{A}$ kernel. Your final line still reads $[U^{ec}(y) - U^{ce}(x)]$; the second must be $U^{ec}(x)$.

5. Divergences — exactly one, and it is collinear

Scanning the reduced integrand over $\{x, x', y, y', z\}$:

no short–distance divergence in any of the 26 collapsing regions — no UV.

region	behaviour	verdict
$z \rightarrow y$ (the $\mathfrak{C}$ kernel)	ρ^{-1} ($\langle I \rangle \rho = 0.5955$ flat)	convergent
$y \rightarrow x$ (the $\mathfrak{A}$ kernel)	ρ^{-1} (0.3900 flat)	convergent — and $[U(y) - U(x)]$ gives another ρ
$ x , x' , y , y' \rightarrow \infty$	falls faster than R^{-2}	convergent
$ z \rightarrow \infty$	flat R^{-2} tail	log divergent

The reason the kernel endpoints give no log is the one that has run through every row: a log needs ρ^{-2} , i.e. the amplitude *and* conjugate kernel singular on the **same** separation. Here they sit on (y, z) versus (y', z) and on (x, y) versus (x', y') — different separations.

5.1 The one divergence, and why it is the fragmentation pole

The z –dependence after the Wilson–line reduction is **only** in the two $\mathfrak{C}$ kernels ($U(z)$ has cancelled), so the z integral is the master

$$T^{mm'} = \int d^2 z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2},$$

whose only divergence is at large $|z|$, where $\langle N^m N'^{m'} \rangle \rightarrow \frac{\delta^{mm'}}{2|z|^2}$. Measuring the coefficient of $\int \frac{d^2 z}{|z|^2}$ directly:

measured	$\left[\frac{c_1+c_3}{2} + c_2\right] \mathcal{X} \text{ (rescaled)}$
−0.896002936	−0.896002936

and symbolically that coefficient is

$$\frac{d_{\perp} p^+}{2S^2} + \frac{p^+}{k^{+2}} + \frac{1}{p^+} = \frac{1}{k^+} \left[\zeta \bar{\zeta} + \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right] = \frac{C_{UV}(\zeta)}{k^+} = \frac{P_{gg}(\zeta)}{2N_c k^+} \quad (d_{\perp} = 2).$$

So the pole's residue is the complete splitting function — all three pieces, including the regular $\zeta \bar{\zeta}$ — with no dim reg needed to see it. And it sits at **large** transverse separation, i.e. small relative transverse momentum: that is a **collinear** divergence, not a UV one. Large separation is where a *fragmentation* collinear region lives (the unobserved daughter collinear to the parent), which together with the $k_{\perp}/\zeta$ hard factor and the $\frac{d\zeta}{\zeta^2}$ measure of §7 makes this unambiguously the fragmentation row.

6. Dimensional regularisation and $\overline{\text{MS}}$

In $d = 2 - 2\epsilon$ (so $\epsilon > 0$ regulates the large- $|z|$ end), Feynman-parametrising and shifting,

$$T^{mm'} = \pi \Gamma(1 + \epsilon) \frac{\Gamma(1 - \epsilon)^2}{\Gamma(2 - 2\epsilon)} (\pi \mu^2 r^2)^{-\epsilon} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right], \quad r = y' - y,$$

$$\frac{C_{\epsilon}}{\epsilon} \longrightarrow \pi \left[\frac{1}{\epsilon} + 2 - \gamma_E - \log(\pi \mu^2 r^2) \right] = \pi (\mathcal{L} + 2), \quad C_{\epsilon} \rightarrow \pi,$$

$$\boxed{\mathcal{L} = \frac{1}{\epsilon_{\overline{\text{MS}}}} - \log(4\pi^2 \mu^2 r^2), \quad \frac{1}{\epsilon_{\overline{\text{MS}}}} \equiv \frac{1}{\epsilon} - \gamma_E + \log 4\pi}$$

(consistent with the cutoff form $\frac{\pi}{2} (\log \frac{R^2}{r^2} + 1) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'}$). Contracting your bracket with $T^{mm'} X^k X^{lk'}$, the two $\frac{2}{k^+}$ pieces cancel ($b_1 + b_3 = \frac{d_{\perp} p^+}{S^2}$ — again $\zeta + \bar{\zeta} = 1$), and:

$$\begin{aligned} \frac{d^3 N}{d^3 k} &= \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{\pi N_c}{k^{+2}} \int_{\Lambda}^{\vee - k^+} \frac{dp^+}{2\pi} \int_{x, x', y, y'} e^{-i \xi \cdot (y' - y)} \left\{ C_{UV}(\zeta) \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\} \\ &\quad \times N_c^{-1} \langle N_c [U(y') - U(x')] [U(y) - U(x)] \rho(x') \rho(x) \rangle \end{aligned}$$

Symbolic difference against the direct ϵ -expansion: **exactly** 0. The $\frac{1}{\epsilon_{\overline{\text{MS}}}}$ inside $\mathcal{L}$, with coefficient $C_{UV}(\zeta) = P_{gg}/2N_c$, is absorbed into the gluon **fragmentation function** $D_{g/g}(\zeta, \mu)$; what is left is $-\log(4\pi^2 \mu^2 r^2)$, the physical μ -dependence that DGLAP evolution of D compensates.

7. The fragmentation variables

$$p^+ = k^+ \frac{1 - \zeta}{\zeta}, \quad dp^+ = -\frac{k^+}{\zeta^2} d\zeta, \quad \int_{\Lambda}^{\vee - k^+} dp^+ = k^+ \int_{k^+/\vee}^{k^+/(k^+ + \Lambda)} \frac{d\zeta}{\zeta^2}$$

The $\frac{d\zeta}{\zeta^2}$ measure, the hard factor at $k_{\perp}/\zeta$, and $P_{gg}(\zeta) \log \mu^2$ are the three hallmarks of a fragmentation function — all three present. The lower limit $\zeta_{\min} = k^+/\vee$ is kinematic; the upper limit $\zeta \rightarrow 1$ is $p^+ \rightarrow \Lambda$.

8. Can you use the $+$ prescription? Yes — here, unlike in the $|\mathfrak{B}_2|^2$ row

The only $\zeta \rightarrow 1$ singularity is $\frac{\zeta}{1-\zeta}$ inside C_{UV} . Write the ζ -integral as $\int d\zeta \frac{\zeta}{1-\zeta} F(\zeta)$ with everything else in F . The $+$ prescription is legitimate iff F is smooth at $\zeta = 1$ with $F(1)$ finite.

After the z integral, the **only** remaining ζ -dependence in F is the phase $e^{-ik \cdot r/\zeta}$:

ζ	F	F'
0.99	-0.145853250	1.715968373
0.999	-0.130531825	1.688830977
0.9999	-0.129013092	1.686130123
1	-0.128844494	1.685830178

Analytic at $\zeta = 1$, and $F(1) = e^{-ik \cdot r} \mathcal{L} \mathcal{X}$ — the **LO** structure. Nothing is deleted and nothing becomes singular.

Contrast with the $|\mathfrak{B}_2|^2$ row, where I told you *not* to apply it naively. There the z integral had not been done, so setting the subtraction point killed the z -phase and manufactured an IR divergence the row did not have. Here the z integral is already done, z is gone, and the subtraction point is harmless. The difference is not the prescription — it is **what is left in F when you evaluate it at the endpoint**.

The identity, verified numerically (agreement $1.6 \cdot 10^{-7}$ at $\delta = 10^{-7}$):

$$\int_a^{1-\delta} d\zeta \frac{\zeta F(\zeta)}{1-\zeta} = \int_a^1 d\zeta \frac{\zeta F(\zeta) - F(1)}{1-\zeta} + F(1) \log \frac{1-a}{\delta}, \quad a = \frac{k^+}{\sqrt{\Lambda}}, \quad \delta = \frac{\Lambda}{k^+}$$

$$\implies \text{isolated log} = F(1) \log \frac{k^+(\sqrt{\Lambda} - k^+)}{\sqrt{\Lambda} \Lambda}.$$

So the $+$ prescription does exactly the right job: it separates a finite, Λ -independent integral from one term proportional to the LO structure times $\log \frac{1}{\Lambda}$ — the piece that goes into the evolution.

Three conditions.

1. **Do the z integral first.** The coefficient of $\frac{1}{\epsilon}$ is ζ -dependent and blows up at $\zeta = 1$; $+$ -subtracting in p^+ before renormalising mixes an endpoint subtraction into an unrenormalised pole.
2. **Do not use Λ and the $+$ prescription for the same singularity.** Once you $+$ -subtract, the Λ -dependence lives *only* in the explicit $F(1) \log \frac{k^+(\sqrt{\Lambda} - k^+)}{\sqrt{\Lambda} \Lambda}$ above. Keeping Λ in the subtracted integral as well double counts.
3. **Your range is $[\zeta_{\min}, 1]$, not $[0, 1]$.** The textbook $\int_0^1 \left[\frac{1}{1-\zeta}\right]_+ d\zeta = 0$ does not hold here; the compensating term is the $\log \frac{1-a}{\delta}$ written above. Either use that form, or extend to $[0, 1]$ and add the $\log(1 - \zeta_{\min})$ back by hand.

9. Remaining bookkeeping

1. After the w, w' integration the measure is still written $\int_{z,w,w'}$ — but w, w' are gone. It should be $\int_z$, and after the z integral, nothing.
2. $\int_{x,y,y'}$ is missing x' : it should be $\int_{x,x',y,y'}$.
3. $d_\perp = 2$ in the final bracket — but **not** before the z integral, where $d_\perp = 2 - 2\epsilon$ multiplies the $\frac{1}{\epsilon}$ and its $O(\epsilon)$ part contributes the finite $\mathcal{X}$ in $\zeta \bar{\zeta}[\mathcal{X} - 2\mathcal{R}]$. Keep it symbolic until §6 is done.
4. The $e^{-ik(y'-y)S/k^+}$ line has a typo: $k+$ should be k^+ .

Checked by `src/row22_check.py`, `row22_colour.py`, `row22_div.py`, `row22_msbar.py`; output in `RESULTS_row22.txt`.